HackResearch 2026

Finding Cancer's Backup Signals

Cancer cells can adapt when a drug blocks part of their signaling system. An alternative signaling route may become more active, potentially allowing the cells to survive. Can we identify these possible backup responses from large-scale measurements of protein phosphorylation?

Kinases are proteins that regulate cellular signaling by adding phosphate groups to other proteins. These modifications, called **phosphorylation**, can change how proteins behave. Drugs that inhibit kinases disrupt these signals, while **phosphoproteomics** measures phosphorylation at thousands of locations across proteins.

This project asks: **When a drug inhibits kinase A, does the resulting phosphorylation pattern suggest increased activity of kinase B?** Such relationships could nominate pairs of kinases for future combination-treatment experiments.

The dataset comes from Hijazi and colleagues' study, *Reconstructing kinase network topologies from phosphoproteomics data reveals cancer-associated rewiring*. The processed MCF7 breast cancer cell dataset contains 9,416 phosphosite entries measured across 61 individual drug treatments, with separate tables of reported phosphorylation changes and corresponding p-values. The study also provides information about which kinases the drugs inhibit and inferred relationships between kinases and downstream phosphorylation sites.

The central research challenge is attribution. One drug can inhibit several kinases, and a phosphorylation site can respond to several signaling pathways. An apparent backup response could therefore reflect biological compensation, overlapping drug targets, or ambiguity in how kinase activity is inferred. The measurements capture responses after one hour of treatment; they do not directly establish long-term resistance or whether a drug combination improves cancer-cell killing.

The aim of this project is an original computational investigation of candidate compensatory kinase relationships. A successful contribution could identify plausible pairs, reveal limits to what the data can establish, or show why an apparently convincing relationship is misleading. The final research write-up should present the question investigated, the evidence obtained, and the uncertainty surrounding its interpretation.

Data and research resources

MCF7 processed dataset - Excel workbook containing the fold and pvalue sheets. Rows identify phosphorylation sites; treatment columns identify drugs. Verify the numerical transformation before interpreting values as absolute fold changes.

Data repository - Supporting inputs, including kinase-inhibitor selectivity information.

Analysis code and documentation - The authors' EBDT method for inferring downstream kinase targets.

Research paper and supplementary datasets - Hijazi et al., Nature Biotechnology 38, 493-502 (2020). Dataset 1: inhibitor selectivity. Dataset 3: inferred kinase-to-phosphosite relationships.